

Chapter 28 — Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

A BRIEF HISTORY AND OVERVIEW

- **1876 — Samuel Gross** coined “**interstitial cystitis**”: chronic bladder inflammation of unknown aetiology.
- **Guy Hunner (1915, 1918)** described “**classical**” **interstitial cystitis (IC)**: bladder pain on filling with cystoscopically inflamed areas of the bladder wall, later termed **Hunner lesions (HLs)**.
- Over the following 50 years, **most patients with bladder pain had no bladder-wall abnormality** on routine cystoscopy.
- **1987 National Institutes of Health/National Institute of Diabetes and Digestive and Kidney Diseases (NIH/NIDDK) criteria**: an expert set of inclusion/exclusion criteria built to give a homogeneous population for clinical trials — **a research definition, not a clinical one**.
- Applied in clinical practice, the NIDDK criteria **excluded about 60% of patients** who would otherwise have been diagnosed clinically. The definition therefore had to broaden, and **bladder pain syndrome (BPS)** is now the most commonly used term; it meets medical pain-taxonomy standards.
- **Current clinical definition** (adopted by the American Urological Association [AUA] Guideline for over a decade): “**an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with lower urinary tract symptoms (LUTS) of more than 6 weeks’ duration, in the absence of infection or other identifiable causes.**”
- **IC/BPS is an umbrella term for a heterogeneous syndrome** — a form of chronic pelvic pain spanning gross inflammatory bladder disease (HLs) at one end and bladder pain with multiple non-urological complaints and a normal-looking bladder at the other.
- **Mounting evidence that BPS is distinct from disease with HLs**. The authors propose “**IC**” for patients with HLs — defined by symptoms plus cystoscopic and histological evidence of visible end-organ inflammatory disease. Some regard IC as a disease distinct from BPS, the latter being a clinical syndrome or symptom complex.
- **US Food and Drug Administration (FDA) meeting**: asked whether IC and BPS should be combined in trials (implying two conditions), it **unanimously chose to study them together** — their symptoms are indistinguishable.
- **Heterogeneity confounds treatment development**; better phenotyping may improve outcomes through **precision-medicine approaches**.
- **Figure 28.1**: urological chronic pelvic pain syndrome (UCPPS) contains IC/BPS (both sexes) and chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS, males). Within IC/BPS, **IC with HLs is a true inflammatory end-organ disease; BPS without HLs is a syndrome or symptom complex**.

Concept — *The history is a progressive widening of the definition from a cystoscopic lesion to a symptom complex; the chapter now narrows it again by splitting off the HL phenotype, because lesion-defined disease and symptom-defined syndrome behave differently.*

DEFINITION AND NOMENCLATURE

- **Definition and name drive everything downstream**: diagnosis, epidemiology, trial protocols, recognition by the profession, naming of support groups, and (in many countries) insurance cover and

reimbursement. Subtle changes in pain character, location, relation to voiding or duration alter all of these.

NAMES IN USE FOR THE CONDITION

Term	Scope and defining features
Interstitial cystitis/bladder pain syndrome (IC/BPS)	Umbrella clinical term; AUA symptom-based definition above
Painful bladder syndrome (PBS)	Synonym
Hypersensitive bladder	Synonym emphasising sensory hypersensitivity
Urological chronic pelvic pain syndrome (UCPPS)	Chronic pain in the pelvic or genital regions. Used by the Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network to encompass IC/BPS (both sexes) and CP/CPPS (males), recognising mechanistic and symptomatic overlap
Chronic primary bladder pain syndrome (International Classification of Diseases, 11th Revision [ICD-11])	Chronic primary pain perceived in the bladder region plus at least one other symptom (pain worse on filling; daytime and/or night-time frequency); not accounted for by infection or by chronic secondary visceral pain

COMPARISON: THE THREE DEFINING FRAMEWORKS

Feature	NIDDK (1987)	AUA clinical definition	ICD-11 chronic primary BPS
Purpose	Research homogeneity	Clinical diagnosis	Classification of chronic pain
Core	Inclusion/exclusion criteria	Pain, pressure or discomfort related to bladder + LUTS	Chronic primary pain in bladder region + ≥ 1 bladder symptom
Duration	—	>6 weeks	Chronic
Exclusions	Strict	Infection or other identifiable cause	Infection or chronic secondary visceral pain
Consequence	Missed about 60% of clinical cases	Broad, inclusive	Places IC/BPS among primary pain conditions

Concept — Every name encodes an assumption about where the pain arises: “cystitis” implies bladder inflammation, “bladder pain syndrome” a symptom complex, “UCPPS” a regional pain disorder, and ICD-11’s “chronic primary” label classes the pain itself as the disease.

EPIDEMIOLOGY

- **Three estimation methods:** self-reported history, symptom identification, clinician diagnosis. **Prevalence estimates rise as the case definition broadens;** methodological differences explain between-study variation.
- **RAND Interstitial Cystitis Epidemiology (RICE) study** — the most sophisticated population-based study: **3.3–7.9 million US females and 2.1–4.6 million US males** have symptoms consistent with IC/BPS. **Mean age of onset 30–40 years.**

Prevalence in Males

- **Female:male ratio 5:1 in clinic-diagnosis studies but 2–3:1 in population studies** — men are under-diagnosed.
- **In men, IC/BPS prevalence exceeds that of CP/CPPS.** Many men labelled CP/CPPS have IC/BPS-type bladder symptoms: in MAPP, most men (largely enrolled as CP/CPPS) reported **“painful bladder filling”**

(pain worsening with filling) and/or **“painful urgency”** (strong urge to void because of pain, pressure or discomfort).

- **Practice: ask men about bladder pain and urinary symptoms, not only genital (penile, testicular, perineal, ejaculatory) pain; re-evaluate for IC/BPS when CP/CPPS treatment fails.** CP/CPPS is covered in Chapter 27.

Prevalence in Children

- **No prevalence data; considered uncommon.**
- **Paediatric voiding dysfunction with bowel and pelvic-floor disturbance is common** and produces pelvic and urethral pain easily confused with IC/BPS.
- **Recall-based studies:** adult female IC/BPS patients report more childhood urgency and recurrent urinary tract infections (UTIs). Those with irritable bowel syndrome (IBS)-like symptoms recalled needing to push to defaecate, voiding only once or twice daily, and painful urination in childhood.
- With the high prevalence of **high-tone pelvic floor dysfunction (HTPFD)** in IC/BPS, this suggests **linked bladder–bowel–pelvic-floor dysfunction dating from childhood.**
- **A small paediatric cohort** with bladder pain, frequency and sensory urgency without infection, evaluated by urodynamics (UDS), cystoscopy and distension, had findings consistent with IC/BPS: **median onset about 5 years; most had long-term remission after bladder distension.** They must be distinguished from dysfunctional voiding.

Prevalence of Hunner Lesions

- **Why HL epidemiology is difficult:** medical codes do not separate HL-positive from HL-negative patients; **most US clinicians do not perform cystoscopy or biopsy;** data come mainly from tertiary or research centres (limited generalisability); and **no pictorial atlas of HLs existed until 2020.**
- **Conventional figure: about 20% of IC/BPS patients have HLs;** reported range among cystoscoped patients 3.5–56% (Table 28.1).
- **HL prevalence rises with age, in both sexes** (Fig. 28.2, two clinical series and the MAPP cohort): significantly higher at **≥50 years than below 50;** uncommon under 30 and present in **over half of patients aged over 60–70.**
- **Progression from non-HL to HL disease has never been documented.**
- Limited studies suggest a **possible genetic susceptibility** to IC/BPS.

TABLE 28.1 PERCENTAGE OF HUNNER LESION PATIENTS WITHIN VARIED INTERSTITIAL CYSTITIS/BLADDER PAIN SYNDROME STUDIES

Group of series	HL prevalence among IC/BPS patients
Early series (1949–1996)	3.5–28%
Interstitial Cystitis Database (1997, US multicentre)	About 10–11% overall; 5.3–33.3% between referral centres
Male-only series (2001, 2004)	10%
Scandinavian tertiary series (2002, 2012)	55–56% — the upper extreme
Other contemporary series (2008–2021)	7.6–39%; most 10–17%

Data from 20 series, 1949–2021 (Hand 1949 to Lai et al. 2021).

- **What the table tests:** HL prevalence is **method- and centre-dependent** — the same disease ranges from a few per cent to over half. High figures come from centres that cystoscope everyone and treat older patients; the working figure is about one in five.

- **Trap:** HLs are present in men too, at a similar rate.

Concept — *Every epidemiological figure depends on which definition and which method were used: symptom-based population surveys find more patients and more men than clinic diagnosis, and HL prevalence depends on whether anyone looked with a cystoscope and how old the patients were.*

NATURAL HISTORY, EFFECTS ON QUALITY OF LIFE, AND ECONOMIC BURDEN

Natural History

- Assessed mainly by **patient recall**. Phenotype differs by age at diagnosis:

Age at diagnosis	Characteristic features
<30 years	More severe urological and non-urological pain, urgency, frequency, dysuria, dyspareunia, genital pain
30–60 years	More chronic overlapping pain conditions (COPCs) such as fibromyalgia syndrome (FMS), widespread pain, somatic symptom burden, depressive symptoms
>60 years	More nocturia, urinary incontinence and HLs

- **Prodromal pelvic pain and irritative voiding often precede diagnosis by decades.** Painless frequency is common among those with prodromes; these patients more often have non-bladder syndromes (e.g. FMS) and a **poorer prognosis** — consistent with the recalled childhood voiding complaints.
- **Mild symptoms at onset → stable at 3 years; chronic fatigue syndrome (CFS) at onset → progression.**
- **MAPP predictors of worsening pain and urinary symptoms at 12 months:** extent of widespread pain, somatic symptom burden and poorer overall health. **Not predictive:** baseline anxiety, depression, general mental health. **Pain catastrophising and perceived stress predicted worse pain.**
- **Follow-up to 9 years: pain and urinary symptoms remarkably stable** — a chronic condition with a stable overall course.
- **Remission is under-captured**, because asymptomatic patients stop attending. In the Interstitial Cystitis Database, about a third had at least one remission of ≥ 3 months; a minority of these lasted ≥ 6 months.
- **Early improvement in longitudinal studies reflects regression to the mean and an intervention effect** (more follow-up and care). Hence the need for **randomised controlled trials (RCTs)**, ideally with a **pre-enrolment run-in**. Regression to the mean, placebo effect and selection bias explain why **treatments promising in uncontrolled series fail in RCTs**.
- **Flares (MAPP):** reported by almost all female UCPPS patients and by every patient meeting IC/BPS criteria. Highly variable in duration (seconds to months), frequency (several a day to once a year or less), type (pain vs urinary) and intensity. **Commonest duration: multiple days;** longer flares bring more pain, urinary symptoms, activity disruption and bother.
- **Case-crossover study: sexual activity was the only consistent trigger of flare onset;** subgroups were triggered by dietary factors and abdominal exercise. **Intervene when a trigger is identified, but no rigid flare-prevention plan.**

Quality of Life

- As with all chronic pain: **decline in mental and physical quality of life (QoL)**, poor coping, anxiety, depression and high stress.
- **Sexual dysfunction in both sexes** profoundly affects QoL — identify and treat it.

- **Catastrophising** (believing the worst will happen and that one could not tolerate it) is common and associated with worse depressive symptoms, mental health, social functioning, vitality and pain; negative affect and catastrophising partly explain disability.
- **Depression and catastrophising (helplessness) predict suicidal ideation.** Cognitive behavioural therapy (CBT) addressing catastrophising and mood may play an important role.
- **Work:** IC/BPS patients are six times more likely than the general population to reduce work time for health reasons, but half as likely as patients with arthritis. In RICE, **the impact of bladder symptoms on life activities was the most consistent predictor of work outcome** → role for behavioural therapy to build coping and self-management.

Abuse and Childhood Trauma

- **Boston Area Community Health (BACH) Survey: emotional, sexual and physical abuse is a risk factor** for IC/BPS-type symptoms, confirmed by case-control and health-maintenance-database studies and by MAPP (more adverse childhood experiences, including sexual and physical violence, than healthy controls).
- **History of sexual abuse** → more pain, fewer voiding symptoms, more widespread pain outside the pelvis; more likely to worsen and less likely to improve over 12 months.
- **Caveat:** data are mostly self-report and recall, so subject to bias. **Draw no conclusion about abuse in an individual**, but remain sensitive to the possibility in all pain patients, and in BPS particularly.
- **Multiple diagnoses → higher rate of previous abuse;** such patients may need referral to a traumatic-stress centre.
- **Neuroimaging: childhood adversity alters brain functional networks and structures** (e.g. the amygdala, which modulates response to threat and stress) — the neurobiological basis of its effect.

Medical Costs

- **Direct costs amount to several thousand US dollars per patient-year**, excluding lost wages, pre-diagnostic costs, alternative therapies and misdiagnosis; Medicare expenditure per beneficiary over 65 is higher still.
- In the first year after diagnosis, **IC/BPS adds substantially to total health-care costs, mostly outpatient; HL patients use more resources than non-HL patients.**

Concept — *IC/BPS is chronic but not progressive: symptoms flare around a stable baseline, so the burden is set less by the bladder than by widespread pain, catastrophising and coping — which is why those, and not baseline mood, predict outcome.*

ETIOLOGY AND PATHOPHYSIOLOGY

- **Unknown and probably multifactorial**, acting through one or more pathways to produce a common symptom set (Fig. 28.3: pain beyond the bladder, regional and widespread; HL-positive = IC, HL-negative = BPS). Many theories; **sparse confirmatory clinical evidence.**
- **Proposed mechanisms:** “leaky” urothelium with defects of the **glycosaminoglycan (GAG) layer**; **mast-cell activation** with histamine release; altered neuroplasticity; neurogenic inflammation; **pelvic organ cross-sensitisation**; dysregulation of the hypothalamic-pituitary-adrenal axis or stress response — possibly combining into a **self-perpetuating process** of chronic bladder pain and voiding dysfunction.
- **IBS, FMS, CFS and other chronic pain disorders may precede or follow IC/BPS** — not inevitably; their causal relationship is unknown.

- **Neural cross-talk in the dorsal root ganglia, spinal cord and brain → central sensitisation and altered functional brain connectivity.** This may explain the **absent physical findings and non-response to local therapy** in many BPS patients — the mechanism is systemic.
- **Infection:** no organism analogous to *Helicobacter pylori* in gastritis has ever been consistently identified, including with microbiome and culture-independent DNA/RNA techniques; negative studies far outnumber positive ones. A **double-blind placebo RCT of 18 weeks of sequential antibiotics** (detailed under Oral Therapies) showed minimal benefit and frequent new non-urinary side-effects. **Active infection is unlikely to drive the process, and antibiotics have no role when urine culture is negative.**
- **Not a primary autoimmune bladder disorder** in the classic sense.
- **Small-fibre polyneuropathy (SFPN):** a newer theory. Found in a proportion of both HL and non-HL patients; bladder capacity and HL status do not differ by SFPN status. **Causal role unknown** — FMS and CFS also show SFPN, and studies are few.

PATHOLOGY

- **No pathognomonic histology** — unsurprising in a syndrome defined by visceral pain; this reinforces the need for phenotyping.
- **Confounders:** disease duration, recent trauma (e.g. hydrodistension), prior bacterial cystitis, and biopsy site and method. Appearances vary widely between patients and between serial biopsies from the same patient.

Biopsy method	Typical finding in IC/BPS
Transurethral resection (TUR) biopsy	Mucosal ruptures, submucosal haemorrhage, mild inflammation
Cold-cup forceps biopsy	Normal in about half

- **Normal biopsy is common in symptomatic patients and does not exclude IC/BPS.** Conversely, inflammatory features appear in 24–76% of non-HL patients. Epithelial denudation, submucosal oedema, congestion and ectasia, and inflammatory infiltrate are more frequent than in controls, yet **over half of IC/BPS specimens are indistinguishable from controls.**
- **IC and BPS differ strikingly in histopathology, chiefly in severity of inflammation.** With **no evidence of temporal transition from BPS to IC**, this argues for distinct phenotypes, perhaps distinct disorders.
- **Classic comparative series:** the **HL group** showed mucosal ulceration and haemorrhage, granulation tissue, intense infiltrate (mainly plasma cells and lymphocytes), raised mast-cell counts and perineural infiltrates. The **non-HL group, despite equally severe symptoms**, had relatively unaltered mucosa with sparse inflammation; the main feature was **multiple small mucosal ruptures and suburothelial haemorrhages (glomerulations)**. Specimens were taken immediately after hydrodistension, so the minimal non-HL findings may be iatrogenic; the same group later found **completely normal biopsies common in non-HL patients.**
- **Severe lymphoplasmacytic infiltrate in almost all HL biopsies versus few non-HL.** In HL patients, **epithelial denudation extends beyond the lesion — a field change** of the bladder surface.
- Most studies find few inflammatory events in non-HL disease, although one reported more bladder-wall fibrosis and mast-cell infiltration.
- **Severely abnormal pathology may predict a poor prognosis** — not consistently.

Mast cells

- **Mast-cell proliferation, activation and degranulation may contribute:** histamine and pro-inflammatory mediators cause pain, hyperaemia and fibrosis — all features of IC/BPS.
- **In IC, mast cells lie predominantly in the detrusor, with a second population in the lamina propria and epithelium** having distinct staining, physiological responses and secretion. This epithelial population is absent in controls and **not involved in BPS**.
- **Mast-cell activation is far more pronounced with HLs**, together with prominent epithelial and detrusor inflammation; in BPS it is sparse → the basic pathological processes may differ.
- **Mast-cell density as a diagnostic marker has failed:** results disparate because of method. Tryptase immunohistochemistry showed higher subepithelial and detrusor density in HL disease but low predictive value, and **non-HL density equalled that of overactive bladder (OAB)**. Digital quantitative analysis: **IC resembles “chronic cystitis”; BPS resembles normal controls**; density tracks overall lymphoplasmacytic density.
- **Mast-cell counts may matter in pathogenesis but have no role in diagnosis.**
- **Role of histopathology: limited to HL patients, to exclude bladder cancer and carcinoma in situ (CIS), eosinophilic cystitis, tuberculous cystitis, ketamine cystitis and any other specific tissue diagnosis.**

Concept — Histology separates the two phenotypes rather than diagnosing the syndrome: *HL disease is a genuine inflammatory lesion with mast-cell activation and a field change, whereas BPS is often histologically normal — so biopsy exists to exclude (CIS above all), not to confirm.*

ASSOCIATED DISORDERS

- **COPCs — IBS, FMS, CFS, migraine, temporomandibular joint disorder (TMJD) — are functional pain syndromes** accounting for part of the **75–80% of patients with pain beyond the bladder**. **Regional pelvic disorders — HTPFD and vulvodynia — often coexist.** These “pain generators” overlap with bladder symptoms, worsen pain and complicate diagnosis and management; the associations also hint at aetiology and aid phenotyping (Fig. 28.4, Table 28.2).
- **Pain distribution: 75–80% have pain beyond the pelvis, 38% widespread in multiple body regions; only 20–25% pelvis-only.** “Pelvic pain and beyond” brings more sensory-type pain, depression, anxiety, stress and sleep disturbance, and worse QoL.
- Key question in widespread pain: **is the bladder the primary disease (less likely) or a bystander — one involved organ in systemic pain?**
- **True somatisation disorder is rare**, despite extrapelvic symptoms.
- **IBS, FMS and CFS are more prevalent than in controls.** In UCPPS, over a third had at least one COPC: **IBS alone was the commonest pattern**, followed by multiple COPCs, with FMS alone and CFS alone less common. COPCs mean more severe and longer-standing symptoms, more depression and anxiety; **multiple COPCs imply a centralised pain component.**
- **IBS — the commonest associated condition** (over 30% of IC/BPS), with the greatest excess over the general population of any functional pain syndrome. IBS patients feel intestinal pain at lower gas volumes than healthy people — **strikingly similar to pain on bladder distension**. **Visceral organ cross-talk**, via convergent sensory pathways in the dorsal root ganglion, spinal cord and/or brain, may explain the link.
- **FMS: chronic widespread non-articular musculoskeletal pain with fatigue, non-refreshing sleep and stiffness.** Shares female predominance, stress triggers, associated symptoms and **response to tricyclic compounds** with IC/BPS. **The “poster child” of centralised (nociceptive) pain — central sensitisation and altered brain pain processing.**

- **Two pain phenotypes: “pelvic pain only” and “pelvic pain and beyond”.** Widespread, possibly centralised pain may not respond when only local therapy is applied → tailor treatment to phenotype.
- **Vulvodynia:** vulvar pain of **at least 3 months** without clear cause; prevalence in female IC/BPS ranges from about one in six to almost all. Shares comorbidities with IC/BPS (IBS, FMS, orofacial pain).
- **Endometriosis:** evaluated by imaging and/or laparoscopy. The concordance of non-infectious inflammatory syndromes in **tissues derived from the embryonic urogenital sinus**, with overlapping functional syndromes, argues for a common cause.
- **Autoimmune disease:** IC/BPS is not a classic primary autoimmune disorder. Crohn disease, systemic lupus erythematosus (SLE), Sjögren syndrome (SS) and rheumatoid arthritis (RA) are infrequent in absolute terms but show an **astoundingly high relative excess**. The most frequent is **inflammatory bowel disease (IBD)/Crohn disease, about 100-fold** the general population (single study, uncorroborated).
- **SLE:** tens-fold higher prevalence; bladder symptoms may predate the diagnosis of SLE. The critical distinction is **association versus lupus cystitis: lupus cystitis runs a more virulent course — gross inflammation, bladder-wall fibrosis, hydroureteronephrosis**. In a few, bladder symptoms arise from **myelopathy of the sacral cord**.
- **SS:** autoimmune exocrinopathy with female preponderance — dry eyes, dry mouth, arthritis, and possibly fever and gastrointestinal (GI) and lung involvement. SS (keratoconjunctivitis sicca with focal lymphocytic sialoadenitis) was found in a substantial minority of IC patients; IC/BPS symptoms occur in up to 5% of SS. **SS patients may have detrusor overactivity (DO)** → evaluate individually before diagnosing IC/BPS.
- **HTFPD: tender, hypertonic pelvic floor and associated muscles in about 80% of IC/BPS.** Higher intravaginal electromyography (EMG) signal correlates with higher IC symptom and problem indices and a shortened pelvic-floor muscle length. Symptom overlap makes both diagnoses harder (Figs. 28.3, 28.5).
- **HTFPD aetiology unknown:** possibly childhood voiding dysfunction; a **guarding response to bladder pain** causing obstructive and irritative voiding; **bidirectional viscerosomatic and viscerovisceral cross-talk**. Resting-state functional magnetic resonance imaging (MRI) shows altered activity and connectivity of pelvic-floor sensorimotor cortical control regions in women with IC/BPS.
- **Phenotype groups** (Fig. 28.4): the **smallest group is bladder symptoms only**; others derive pain from regional (HTFPD, vulvodynia, endometriosis, IBS) or widespread (FMS, CFS, migraine, TMJD) syndromes, often several; **autoimmune/immune-mediated disease (SLE, RA, IBD) may form a separate group. No evidence that IC/BPS progresses from bladder-centric to regional to widespread pain.** Heterogeneity supports phenotyping for precision treatment.

TABLE 28.2 DIAGNOSES FREQUENTLY REPORTED IN INTERSTITIAL CYSTITIS/BLADDER PAIN SYNDROME PATIENTS

Condition	Group	Excess over general population
IBS	Regional (visceral)	Marked; commonest associated condition , also in the contemporary MAPP cohort
Vulvodynia	Regional	Marked on self-report
FMS	Widespread	Marked
CFS	Widespread	Modest to marked
Migraine; TMJD; low back pain	Widespread / other	Modest (about two- to three-fold)
Allergy; sensitive skin; asthma	Allergic	Modest; allergy itself very common
IBD	Autoimmune	Very large (about 100-fold), single study

Condition	Group	Excess over general population
SLE	Autoimmune	Very large (tens-fold)
Sjögren syndrome; RA	Autoimmune	Large (about 10- to 15-fold)
More than one COPC	—	A minority of UCPPS in MAPP

Data from van de Merwe (2003, 2007), Alagiri et al. (1997), Peeker et al. (2003), Nickel et al. (2010) and Krieger et al. (2015; MAPP UCPPS cohort). The general-population prevalence of vulvodynia is now estimated at 8% (formerly 15%).

- **What the table tests:** the difference between **commonest** and **most over-represented**. Functional syndromes (IBS above all) are common in IC/BPS; autoimmune diseases are rare in absolute terms but show the largest relative excess.
- **Trap:** a bladder picture in SLE may be lupus cystitis, with upper-tract damage, not IC/BPS.

Concept — *The bladder is often one of several pain generators rather than the source; how much pain is pelvis-only versus widespread determines whether bladder-directed therapy can succeed, so associated disorders are read as phenotype markers, not comorbid noise.*

DIAGNOSIS

Symptoms

- Clinical syndrome per the AUA definition; the wording deliberately includes **pressure and discomfort** as well as pain, because patients voice them.
- **Pain typically suprapubic and worse with filling (“painful bladder filling”)**, temporarily relieved by emptying → frequent voiding day and night to relieve pain.
- **Dyspareunia common; isolated dysuria or urethral pain uncommon.**
- Associated LUTS: frequency, nocturia, “urgency”. The International Continence Society (ICS) urgency definition — “**a sudden compelling desire to pass urine, which is difficult to defer**” — suits OAB but is poorly understood by patients and clinicians. The overlap has prompted the view that IC/BPS and **OAB-dry** (urgency without urgency urinary incontinence [UUI]) form a **spectrum of bladder hypersensitivity**. Qualitative differences remain (Fig. 28.5):

Feature	OAB	IC/BPS	HTPFD
Urge	Sudden	Strong, persistent or constant; mounting, not sudden	Persistent urge
Driver	Fear of leakage	Pain, pressure, discomfort with filling (“painful urgency”)	Pelvic pressure
Pain	Absent	Present	Present
Incontinence	UUI	Not a feature	—
Voiding	—	Frequency	Obstructive voiding

- **Pain and UUI are the discriminators** between IC/BPS and OAB.

Basic Evaluation

- **The hardest part is excluding other identifiable causes of pelvic pain;** in the authors’ experience, **failure to diagnose other pain generators or confusable disease is among the commonest reasons for treatment failure.** IC/BPS is a clinical diagnosis of exclusion (Table 28.3).

- **Identify and treat all pelvic pain generators, usually stepwise** — the principle of the AUA-derived diagnostic algorithm (Fig. 28.6).
- **History, examination, post-void residual (PVR) and urine studies suffice in almost all uncomplicated presentations.**
- **History and examination** as in Box 28.1 — above all, **ask about ketamine**, palpate the pelvic floor for HTPFD, exclude vaginitis and pelvic mass, and perform a **cotton-swab (Q-tip) vulvar test for vulvodynia** in women and genital examination with **digital rectal examination (DRE)** in men.
- **AUA 2022 basic studies: PVR (bladder scanner), urinalysis, urine culture. Urine cytology is considered when there are risk factors for urothelial malignancy** — smoking history, degree of haematuria, age >40. Microscopic haematuria is evaluated per the AUA/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) Microhematuria Guideline (2020).
- **Symptom instruments:** O’Leary–Sant Interstitial Cystitis Symptom and Problem Index (ICSI/ICPI); Genitourinary Pain Index (GUPI, male and female); Bladder Pain/Interstitial Cystitis Symptom Score (BPIC-SS); pain visual analogue scale (VAS) 0–10; Global Response Assessment (GRA).
- **Score pain and urinary symptoms separately — not as one composite** (as ICSI/ICPI does), because an individual’s pain and urinary trajectories diverge over time and after treatment.
- **Widespread pain: self-administered body map** (Fig. 28.8) — **widespread = 3 or more of 12 regions in addition to the pelvis.**
- **Male CP/CPPS overlap** is so striking that MAPP studies the two together as UCPPS. Of the confusable conditions, two merit special mention: **ketamine cystitis and HTPFD.**

TABLE 28.3 DISORDERS THAT MAY BE CONFUSED FOR INTERSTITIAL CYSTITIS/BLADDER PAIN SYNDROME

Group	Confusable disorder	Excluded or diagnosed by
Malignancy	Bladder cancer and CIS	Cystoscopy, biopsy
	Prostate cancer	Examination, serum prostate-specific antigen (PSA)
	Cervical, uterine, ovarian cancer; colon cancer	Bimanual examination, pelvic ultrasonography; colonoscopy
Infection	Common intestinal bacteria	Routine culture
	<i>Chlamydia trachomatis</i> , <i>Ureaplasma urealyticum</i> , <i>Mycoplasma hominis</i> , <i>M. genitalium</i> , <i>Corynebacterium urealyticum</i> , <i>Candida</i> spp., <i>Mycobacterium tuberculosis</i>	Special cultures; dipstick — “sterile” pyuria → culture for <i>M. tuberculosis</i>
	Herpes simplex, herpes zoster; vaginal candidiasis	History, examination; vaginal cultures
	Chronic bacterial and non-bacterial prostatitis	History, examination, culture
Iatrogenic, drugs	Radiation; chemotherapy including cyclophosphamide; tiaprofenic acid	History
	Ketamine cystitis	History, cystoscopy, upper-tract changes (hydronephrosis)
Outflow, emptying	Bladder neck obstruction, neurogenic outlet obstruction	History, UDS — videourodynamics for bladder neck obstruction
	Benign prostatic obstruction	History, examination, uroflowmetry and pressure–flow in selected cases

Group	Confusable disorder	Excluded or diagnosed by
Functional bladder	Incomplete emptying (retention)	Ultrasound PVR
	OAB	Symptom specifics (Fig. 28.5); UDS as appropriate
Structural	Bladder stone; lower ureteric stone	Imaging (computed tomography [CT], bladder or transvaginal ultrasonography) or cystoscopy
	Urethral diverticulum	History, examination, urethral MRI
Bowel	Vaginal prolapse; abdominal wall hernia	History, examination
	Diverticulitis, IBS, IBD	History, examination, colonoscopy
Gynaecological	Endometriosis, adenomyosis	History, examination, MRI/pelvic ultrasonography or diagnostic laparoscopy
	Pelvic congestion syndrome	History, pelvic ultrasonography
	Vaginal atrophy	Menopause history, oestrogen use, examination
	Vulvodynia	History, Q-tip examination of vulva
Neuromuscular	Pudendal nerve entrapment	History, examination; nerve block may prove diagnosis
	Pelvic-floor muscle-related pain	History, pelvic-floor examination

Modified from the European Society for the Study of Interstitial Cystitis (ESSIC) proposal (van de Merwe et al., 2008).

- **What the table tests:** almost every exclusion is made by **history and examination**; specific tests are reserved for specific suspicions (cystoscopy for cancer, special cultures for sterile pyuria, videourodynamics for bladder neck obstruction, nerve block for pudendal entrapment).
- **Traps:** sterile pyuria means tuberculosis until excluded; ketamine cystitis is flagged by upper-tract change, which IC/BPS does not produce.

BOX 28.1 HIGHLIGHTS OF CLINICAL EVALUATION FOR THE PELVIC PAIN PATIENT

- **History — the pain:** location and relationship to filling and emptying (painful filling, relief after voiding); initiating event; duration, quality, radiation; regional pain (penile, urethral/dysuria, vulvar, perineal, perianal, testicular, inguinal, coccygeal, abdominal, lower back); **painful urgency**; relation to menstrual cycle; sexual pain (ejaculation, orgasm, entry, deep dyspareunia); triggers and relievers (e.g. diet); response to previous therapy.
- **History — function and background:** voiding and/or bowel dysfunction; urological, gastroenterological, gynaecological, neurological and rheumatic disease; previous bladder or pelvic surgery; previous UTI; pelvic malignancy and its treatment; **pelvic irradiation; cyclophosphamide; ketamine**; autoimmune disease; associated syndromes (IBS, FMS, CFS, vulvodynia, endometriosis, migraine, TMJD). **Rule out other causes — IC/BPS is a diagnosis of exclusion.**
- **Examination:** abdomen (suprapubic tenderness, tender abdominal-wall muscles, hernia, distension, bladder distension). **Women:** discharge, urethral diverticulum, caruncle, mesh complication, atrophy, vulvar dermatoses (introital atrophy, lichen sclerosus); Q-tip vulvar pain mapping; one-digit vaginal palpation of bladder, bladder neck and urethra; tenderness, hypertonicity or trigger points of pelvic floor and obturator internus; bimanual examination. **Men:** DRE with pain mapping of the scrotal–anal region; tenderness, hypertonicity or trigger points of anal sphincter, pelvic floor and obturator internus. Focused neurological examination.
- **Laboratory:** urinalysis; urine culture; cytology in risk groups.

- **Symptom evaluation:** frequency–volume chart; numeric pain rating (VAS); symptom score (ICSI/ICPI, GUPI, BPIC-SS, GRA); body map.
- **Other evaluations:** **office flexible cystoscopy over 50** (diagnostic value of hydrodistension under anaesthesia controversial); bladder biopsy if a lesion is found (e.g. to exclude CIS); local anaesthetic challenge; UDS (optional); haematuria per AUA guideline.

Ketamine Cystitis

- **Ketamine: N-methyl-D-aspartate (NMDA) receptor antagonist;** anaesthetic and newly popular antidepressant. Recreationally a “club drug” — rapid onset, short action, and **dissociative anaesthesia** (unconscious, trance-like state with profound analgesia); many street names (e.g. “special K”, “ket”, “vitamin K”).
- Recreational use began in Asia and spread internationally; substantial US lifetime use, and common among young club-goers (slightly more in men).
- **Ketamine and/or its urinary metabolites (e.g. hydroquinone) probably cause the pathology.**
- **Clinically almost indistinguishable from IC/BPS.** Early disease may be LUTS alone; **progression can be rapid after about a year of use:** dysuria, worsening frequency, **gross haematuria**, pelvic pain. Severity of symptoms and inflammation tracks **duration, frequency and dose.**
- Continued use → **gross inflammation resembling HLs but more widespread** → irreversible **bladder-wall fibrosis, hydronephrosis (about 30%) and renal insufficiency (about 10%).**
- **Management: early identification and cessation** — the most important and hardest part; **rehabilitation is essential;** cessation improves LUTS.
- **A history of ketamine use is strongly suggestive, but often not volunteered.** Be especially wary of the young patient (<30) with an IC/BPS picture and visible bladder-wall inflammation.

High-Tone Pelvic Floor Dysfunction

- **Synonyms:** pelvic-floor hypertonicity, non-relaxing pelvic floor dysfunction, pelvic floor spasm, pelvic floor myalgia. Common across urological, gynaecological and colorectal pain syndromes.
- **Symptoms:** pelvic pressure, persistent urge, sensation of incomplete emptying (poor emptying in severe cases), **obstructive voiding** (hesitancy, weak stream), constipation, **dyspareunia (often felt the next day)**, perineal, penile and ejaculatory pain. Storage symptoms may be mistaken for OAB.
- **UDS not essential:** shows voiding dysfunction; **video-UDS shows poor relaxation of the external sphincter.**
- **Suspect HTPFD with dysfunctional voiding + bowel disturbance + sexual pain**, often dating from childhood.
- **Hallmarks: hypertonic pelvic and adjacent muscles, muscle banding, myofascial trigger points** — tender “knots” in a taut band that reproduce pain. Tender, hypertonic pelvic floor in about 80% of IC/BPS; trigger points in a similar proportion, multiple in most.
- **Standardised six-site examination** (Fig. 28.7): left and right anterior levator ani (pubococcygeus) → left and right posterior levator ani (iliococcygeus) → left and right obturator internus (lateral). **Women vaginally in lithotomy; men transrectally, bending over.**
- **HTPFD magnifies IC/BPS pain or is the sole cause;** missing it commonly produces a suboptimal response, because the treatments differ → refer for evaluation by a pelvic-floor physiotherapist.
- **Multimodal programme:** physical therapy; behavioural modification (avoid straining, “reverse Kegels”, stress reduction); topical heat; skeletal-muscle relaxants; control of constipation; anaesthetic or onabotulinumtoxinA (BTX-A) trigger-point injections.

Beyond the History, Physical Examination and Urine Studies

Frequency Volume Chart

- **One 24-hour chart of times and volumes usually suffices**, for diagnosis and follow-up.
- IC/BPS shows frequent low-volume voids with a narrower range than OAB, **but the diary cannot distinguish IC/BPS from OAB**.
- **Pelvic pain with infrequent or high-volume voids (or high cystometric capacity) → search for a non-bladder source of pain.**

Urodynamics

- **AUA: not part of the standard evaluation in uncomplicated cases.** Its value is to identify other functional pathology that confounds diagnosis. Useful when **empirical therapy fails, the presentation is complicated or confusing** (pain with significant incontinence, high PVR); complex cases may need full video-UDS.
- **Cystometry in IC/BPS: normal detrusor function, reduced capacity, hypersensitivity** (strong desire to void at low volume); **compliance normal**, because hypersensitivity prevents filling to the point of non-compliance. **Pain on filling that reproduces symptoms is very suggestive.** An end-stage fibrotic bladder has reduced capacity and compliance.
- **Uroflow or pressure-flow in men** to evaluate outlet obstruction. **Bladder outlet obstruction (BOO) is common in women with IC/BPS**, presumably from pelvic-floor dysfunction; typical voiding-dysfunction symptoms merit dedicated therapy, and UDS adds little that changes care.
- **DO coexists in 15–19%; DO on UDS does not exclude IC/BPS.** Using UDS to exclude DO as a cause of storage symptoms (not pain) is debatable: empirical treatment is justified first-line, and **amitriptyline's anticholinergic action** helps storage symptoms anyway.

Office Cystoscopy

- **AUA 2022 change: cystoscopy for patients in whom HLs are suspected — specifically those over 50**, where prevalence is higher. Rationale: easy cystoscopic recognition, coupled with **therapies specific and unique to this phenotype**.
- **The only way to identify HLs is direct visualisation.** HL extent correlates with symptom severity and capacity, but **HLs cannot be predicted from the clinical presentation or urine studies**.
- **Office flexible cystoscopy:** minimal discomfort; often immediate reassurance about bladder cancer; **most HLs visible without formal hydrodistension under anaesthesia.** A 2020 pictorial atlas aids recognition (Fig. 28.9).
- Some patients prefer anaesthesia; **gentle distension under anaesthesia may reveal additional HLs**.
- **Appearance is variable:** most show a **classic stellate pattern — vessels radiating from a central area of erythema or coagulum with inflammation**; some are blanched and non-inflamed with a white stellate look; lesions may **bleed with filling as a red “waterfall”**, disrupting centrally; **fibrous bridging between lesions further restricts capacity**.
- **Map lesions early; avoid overdistension**, which causes bleeding that obscures them. **Lesion suspicious for CIS → biopsy.** HLs rarely occur at the trigone — **biopsy any trigonal lesion**. Narrow-band imaging may reveal subtle lesions.
- **Also consider cystoscopy** after failed conventional therapy, with a complicated presentation or previous abnormal cystoscopy; **offer it** to exclude stones, cancer, CIS or suspected BOO, and as part of haematuria work-up.

BOX 28.2 CLINICAL PEARLS FOR OFFICE CYSTOSCOPY

- **Filling assesses capacity** (best measured from the voiding diary) and **reproduces filling pain**.
- **The scope tip is an examining finger**: identifies hyperalgesic sites (e.g. bladder neck); “touching” an HL reproduces the patient’s pain.
- **Drain and instil anaesthetic** (e.g. alkalinised lidocaine): significant relief suggests bladder origin; no relief does not exclude IC/BPS.

Local Anaesthetic Challenge

- **Intravesical local anaesthetic localises pain to the bladder**, as opposed to adjacent generators (pelvic floor, GI, gynaecological); performed after cystoscopy or as a stand-alone office test.
- **Authors’ mixture: 15 mL 0.5% bupivacaine + 15 mL 2% lidocaine (30 mL)**; other lidocaine ± bupivacaine combinations acceptable.
- **Improvement → bladder-centric pain; no response does not exclude IC/BPS**. Responders may be offered intravesical therapy.

Potassium Sensitivity Test

- **Intravesical potassium chloride (KCl) challenge**: compares the sensory-provocative effect of sodium with that of **0.4 M KCl**; pain or symptom provocation = positive. Whether a positive result reflects **epithelial permeability or sensory hypersensitivity** is unclear. It was meant to diagnose IC/BPS without cystoscopy and hydrodistension.
- **Not recommended; perform a local anaesthetic challenge instead**. With a symptom-based diagnosis, a painful provocative test is no longer needed.
- **Neither sensitive nor specific**: negative in up to a quarter of patients meeting strict NIDDK criteria; positive in about a third of asymptomatic men and of a fixed population of textile workers; positive in a quarter with OAB, virtually all with radiation or acute bacterial cystitis, and most men with CP/CPPS; non-specific in gynaecological chronic pelvic pain and urethral syndrome. **Does not predict response to oral pentosan polysulfate (PPS)**.

	Potassium sensitivity test	Local anaesthetic challenge
Principle	Provokes pain	Relieves pain
Tolerability	Painful	Well tolerated
Accuracy	Neither sensitive nor specific	Positive localises to bladder; negative does not exclude
Predicts treatment	No (not PPS)	Yes — selects intravesical therapy
Status	Not recommended	Recommended alternative

Cystoscopy With Hydrodistention and Biopsy as a Diagnostic Procedure

- **Cystoscopy with hydrodistension (CHD) was therapeutic long before it was diagnostic**. Under anaesthesia, sufficient distension reveals **glomerulations** — submucosal pinpoint petechial haemorrhages (Fig. 28.10A).
- **Sequence**: fill to **80 cm H₂O for 1–3 minutes** → drain (terminal effluent often blood-tinged) → refill → reinspect: glomerulations develop throughout the bladder and may not be seen without anaesthesia. Mucosal tears may develop, often near the bladder neck (Fig. 28.10B).
- **Glomerulations are not specific**: seen after radiotherapy, with carcinoma, after toxic chemicals or chemotherapy, often in **dialysis patients or after urinary diversion** (bladder unfilled for long periods), in most men with prostate pain syndromes, in up to a fifth of men undergoing transurethral resection of the prostate (TURP) for LUTS, and **in normal individuals after overdistension**. Conversely, **IC/BPS**

may show no glomerulations or a normal bladder under anaesthesia, and their number does not correlate with severity. They may simply reflect **distension after chronic underfilling because of pain**.

- **Hydrodistension to provoke HLs has been questioned:** most HLs are seen at office cystoscopy, and a lesion visible only after distension far beyond physiological capacity may represent a different point in the spectrum or a different phenotype.
- **Suspicious lesion → biopsy to exclude cancer or CIS. Cold-cup biopsy after, not before, hydrodistension** — to avoid perforation. Keep a **low threshold for cystoscopy in patients at higher risk of malignancy** (smoking, haematuria): case series document bladder cancer and CIS misdiagnosed as IC/BPS or CP/CPPS.
- **Diagnostic value of CHD is clinically limited**, despite a possible therapeutic role. **Biopsy only to exclude disorders suggested by the appearance — commonly to differentiate HL from CIS.**

Concept — *Diagnosis is exclusion plus localisation: symptoms define the syndrome, exclusions keep it honest, and each further test — pelvic-floor palpation, body map, cystoscopy over 50, anaesthetic challenge — exists to locate the pain generator that will be treated, not to prove IC/BPS.*

MAPPING CLINICAL PHENOTYPES OF IC/BPS TO TREATMENT STRATEGIES

- Similar symptoms, heterogeneous population → **no single treatment can help the majority**, because each mechanism addresses one phenotype. In trials, a **real benefit in one phenotype is diluted in all-comers, producing an overall negative trial**. Better phenotyping should improve outcomes and permit precision treatment; there is evidence that phenotypes respond preferentially to specific treatments.
- **Four clinically identifiable phenotypes** (Fig. 28.11):

Phenotype	Identified by	Pain centre	Treatment
Hunner lesion	Cystoscopy (offered over 50)	Bladder — true end-organ inflammation	Fulguration and/or triamcinolone injection; refractory → ciclosporin A
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Bladder	Intravesical therapies
Pelvic-floor tenderness/HTPFD	Standardised pelvic-floor examination	Pelvic floor	Myofascial physical therapy
Widespread pain	Body map; COPCs	Central (centralised pain)	Systemic: tricyclic antidepressants (TCAs), gabapentinoids, management of anxiety, depression and catastrophising, CBT, multidisciplinary care

Hunner Lesion Phenotype

- Distinct in demographics, pathology and urinary markers, yet **identified only by cystoscopy** (office flexible, without hydrodistension); **bladder-centric, true end-organ inflammation**.

Pelvic Floor Tenderness/High-Tone Pelvic Floor Dysfunction Phenotype

- Positive standardised examination (tender hypertonic muscle, banding, pain-reproducing trigger points) → **pelvic-floor-centric phenotype; critical to identify**, because level I evidence directs it to myofascial physical therapy.

Widespread Pain Phenotype

- **Over a third of IC/BPS patients have widespread pain** outside the pelvis; easily identified with an **office body map**. Often other COPCs: IBS, FMS, CFS, chronic migraine, TMJD, endometriosis, vulvodynia, chronic low back pain.
- **Centralised pain → inadequate response to local therapy alone** (intravesical instillation, pelvic-floor physiotherapy, PPS). More psychosocial difficulty, and **altered grey-matter volume and functional connectivity on MRI** → needs systemic therapy targeting centralised pain.
- **MAPP Network reanalysis (2023) of two NIH RCTs** — hydroxyzine and/or PPS vs placebo, and amitriptyline vs placebo — both negative in all-comers. Stratified by pain phenotype, significant effects emerged:

Phenotype	Hydroxyzine / PPS vs placebo	Amitriptyline vs placebo
Localised pelvic pain	Significant improvement in pain and urinary symptoms	No benefit
Widespread pain	No benefit	Significant improvement in pain and urinary symptoms

- Localised pain responds to bladder/pelvis-centric drugs; widespread pain to systemic drugs. **Requires prospective validation**, but illustrates phenotype-driven daily management.

Other Bladder-Centric Features

- Intravesical therapy is directed at **bladder-centric pain**; temporary relief with an anaesthetic challenge makes intravesical anaesthetic a viable treatment.
- **Low anaesthetic bladder capacity (<400 mL) at hydrodistension** → worse pain and frequency, more bladder inflammation, fewer reports of depression, IBS, endometriosis and systemic pain — a more bladder-centric picture.
- But hydrodistension is invasive, needs general anaesthesia, and neither technique nor capacity measurement is standardised → **value in phenotyping currently unknown**; how to treat <400 mL differently from ≥400 mL is unclear. Many with <400 mL have HLs and many with ≥400 mL have systemic pain, so **the categories overlap with the HL and widespread-pain phenotypes**.

Concept — **Treatment follows the pain generator**: *bladder-centric disease gets bladder-directed therapy, pelvic-floor-centric disease physiotherapy, and centralised pain systemic therapy* — and a trial negative in all-comers can conceal a positive result in the right phenotype.

TREATMENT OF INTERSTITIAL CYSTITIS/BLADDER PAIN SYNDROME

General Principles of Therapy

Setting Expectations

- **No “home run” therapy exists** — a consequence of heterogeneity and unknown aetiology. Few treatments give significant durable benefit to most patients, leaving a trial-and-error approach that breeds frustration.
- Frustration and secondary depression are reduced by **setting goals and realistic expectations, working as partners, and reassessing periodically**. **No single treatment cures IC/BPS**; the reasonable goal is acceptable QoL while coping, not complete remission.

- Most patients have **persistent symptoms with a stable course and intermittent flares**; balance tolerating residual symptoms against pursuing invasive therapies of uncertain added benefit. Build a trusting relationship; **agree a flare plan** (e.g. anaesthetic instillation) to reduce stress and uncertainty.

Shared Decision Making

- Decide after informing the patient of risks, benefits and alternatives; allow latitude in choosing treatments.
- **Conservative first, escalating for poor symptom control and unacceptable QoL** (Fig. 28.6). **Initial aggressive or multimodal therapy** (behavioural + oral + intravesical) is appropriate for severe symptoms unlikely to respond to conservative measures; concomitant treatments address different aspects of the symptom profile and may reduce the need for aggressive options.
- **Placebo effect and regression to the mean are common in the first 3–4 weeks**. If not improving → re-evaluate for other pain sources or misdiagnosis.
- **Stop ineffective treatments after a fair trial**, and discuss stopping long-standing therapy of doubtful benefit, even temporarily.

Multidisciplinary Approach

- **CBT** from experienced mental-health professionals for catastrophising.
- **Suspected pudendal neuralgia → pudendal nerve blocks** by trained clinicians — therapeutic and diagnostic.
- **Anaesthetic or BTX-A trigger-point injections** (BTX-A not FDA-approved for this) when directed physical therapy has plateaued.
- **Chronic opioid therapy is not routinely recommended**; carefully selected patients may benefit under pain specialists using “universal precautions”.
- **Early gynaecological, gastroenterological and rheumatological consultation**, especially with widespread pain or COPCs (endometriosis, vulvodynia, IBS, FMS, CFS).

Overview of Treatment Algorithm

- **AUA 2022 update**: rule out confusable conditions (UTI, OAB, vulvodynia); complicated presentations may need extra tests (e.g. UDS).
- **HL → fulguration and/or triamcinolone injection (Guideline Recommendation)**; **refractory HL → trial of oral ciclosporin A (CyA)**.
- **Pelvic-floor tenderness/HTPFD → myofascial physical therapy (Guideline Standard)**.
- **No longer first- to sixth-line tiers**; treatments are grouped as behavioural/non-pharmacological, oral, intravesical, procedures and major surgery → individualised care with shared decision-making. **Patients may move to more aggressive therapy (except major surgery) or multimodal therapy without first failing “lower-tier” options.**

Modality	Advantage	Drawback
Behavioural/non-pharmacological	Non-invasive, low risk	Needs a highly motivated patient
Oral	Convenient	Systemic side-effects
Intravesical	End-organ directed; no systemic side-effects	Catheter anxiety and discomfort; frequent visits or self-instillation
Procedures	More durable relief	Invasive
Multimodal	Avoids adverse effects of high-dose single drugs	—

Treatments for Interstitial Cystitis Patients With Hunner Lesions

- Systematic review and meta-analysis: HL and non-HL patients differ in demographics, presentation, urinary markers and treatment response. **Only cystoscopy identifies HLs**; most are seen at office cystoscopy.
- **AUA Recommendation: fulguration and/or triamcinolone injection into the lesions.** Biopsy at the surgeon's discretion if never biopsied, to exclude CIS.
- Patients declining anaesthesia may have behavioural, oral and/or intravesical therapy after shared decision-making. **HL patients respond as well as non-HL patients to amitriptyline, dimethyl sulfoxide (DMSO), BTX-A and sacral neuromodulation (SNM).**
- **Most HL patients (75–90%) improve significantly after fulguration and/or triamcinolone.** Mean interval to re-treatment **7–12 months after triamcinolone, 11–29 months after fulguration.** Recurrence after months is expected → **counsel that periodic re-treatment may be needed.**
- **Effect of repeated treatment on capacity is uncertain:** one series showed no fall with repeated fulguration; another showed about a 20% fall after repeated fulguration or resection combined with hydrodistension.
- **TUR of HLs** relieves symptoms (initial disappearance of pain in all in one series), but carries **perforation risk with anterior lesions.**
- **Neodymium:yttrium–aluminium–garnet (Nd:YAG) laser** gives similar results, but **forward scatter makes bowel injury an ever-present danger**; with fulguration/triamcinolone safe and effective, lasers are hard to justify.
- **Energy applied to glomerulations or normal-looking tissue does not help.**

BOX 28.3 CLINICAL PEARLS OF HUNNER LESIONS TREATMENT

- **Sequence:** under anaesthesia (the procedure is painful) → on entry, before distension bleeding, note **HL location, number and size** (HLs are visible without formal hydrodistension) → mild distension elicits bleeding towards lesion centres → biopsy suspicious lesions; **biopsy any trigonal lesion** (HLs rare there) → avoid overdistension; **fulgurate at low volume** → **triamcinolone submucosally at 2–3 mm depth in 0.5-mL aliquots**, either undiluted 40 mg/mL or diluted 12 mg/mL mixed 1:1 with 0.5% bupivacaine and 2% lidocaine → lesions are friable and bleed after injection, so fulguration usually follows → empty and instil a 1:1 mix of **20 mL 1% lidocaine and 20 mL 0.25% bupivacaine** before recovery.
- **Fulguration technique:** **outline the outer margin with the Bugbee electrode, then work towards the centre** — not outward, because erythema and oedema keep expanding peripherally as inflammation propagates with cautery.

Ciclosporin A for refractory Hunner lesions

- **Oral CyA**, an immunosuppressant/immunomodulator, is an option for HL disease refractory to fulguration and/or triamcinolone.
- Early trial: frequency fell, voided volumes rose, pain fell or disappeared; **symptoms recurred on stopping.** Long-term follow-up (mean about 5 years): most refractory IC patients became pain-free and **capacity more than doubled**; relapse after stopping responded to restarting.
- **CyA far superior to PPS at 6 months** in an open-label RCT. Multicentre retrospective data: success in most HL patients but few non-HL → **HL is the target subgroup**; judge on a **3–4-month trial.** Similar results in patients failing two prior therapies, and most HL patients in a recent series were moderately or markedly better.
- **Side-effects in 30–55%**, may force discontinuation: **nephrotoxicity, hypertension**, immunosuppression and infection, lymphoma, gingival hyperplasia, alopecia, tremor; many drug and food interactions.

- **Dose 1–3 mg/kg/day in two divided doses — lower than transplant doses;** titrate to the lowest effective dose. A **2-hour post-dose serum level** aids monitoring and allowed dose reduction in about 40%.
- **Monitor renal function and blood pressure closely,** plus hepatotoxicity, hyperuricaemia, hypomagnesaemia and malignancy (skin cancer, lymphoma): full blood count, renal and hepatic panels, magnesium and urate — e.g. at 1 month, then 3-monthly.

Treatments for IC/BPS Patients With Pelvic Floor Tenderness/High-Tone Pelvic Floor Dysfunction

- Biofeedback, soft-tissue massage and other physical therapies relax the pelvic floor — reasonable given the strong HTPFD association.
- **AUA Guideline Standard: offer appropriate manual physical therapy to patients with pelvic-floor tenderness, if trained therapists are available** (Figs. 28.6, 28.11).
- **Landmark multicentre RCT (FitzGerald et al., 2012): 10 sessions of myofascial physical therapy vs global therapeutic massage → moderate or marked improvement at 3 months in 59% vs 26%.**
- **Goals:** relax pelvic-floor, abdominal and hip trigger points; lengthen contractures; release myofascial and connective-tissue restrictions and hypertonicity — many tender patients show hypertonicity or non-relaxation.
- **Avoid pelvic-floor strengthening (Kegel) exercises — they may exacerbate pain.**

Conservative Therapies

- **Minimal risk, so an intuitive start:** stress reduction, exercise, warm baths, maintaining a normal lifestyle. In a large patient survey, dietary change, fluid management, heat or cold and stress reduction helped most respondents.
- **Stress worsens pain and urgency in IC but not in controls;** acute stressors exacerbate symptoms. Mindfulness-based stress reduction is promising; yoga, meditation and relaxation techniques (deep breathing) may help; maladaptive coping worsens symptoms. **Catastrophising — the irrational, consuming fear of disaster — amplifies pain and is modifiable with CBT.** A small RCT found guided imagery better than rest.
- Patients learn to avoid their own triggers. **Non-dietary flare triggers: UTIs, stress, tight clothing.** Self-help for flares: heat or cold to abdomen/pelvis, sitz bath, meditation, visualisation, hypnosis, rest.
- **Acupuncture: disappointing** — mostly no better than placebo, sham or standard care. An open-label study improved multiple parameters for 3 months, but symptoms returned to baseline by 1 year (placebo and bias). A small single-blind RCT found electro-acupuncture no better than minimal acupuncture → needs sham-controlled trials with standardised needle placement.

Diet

- **Dietary triggers are reported by a majority of patients (53–93%);** surveys are methodologically weak but **remarkably consistent: caffeine, alcohol, artificial sweeteners, hot peppers and acidic drinks such as cranberry juice.**
- **Evidence conflicts:** a small placebo-controlled study found no diet–symptom relationship; a Japanese controlled trial of intensive dietary manipulation (restricting tomatoes, soy and tofu, spices, excess potassium, citrus and acid-inducing foods) improved symptoms up to 1 year; a **case-crossover study linked flares to tomatoes, yoghurt, spicy food, alcohol and caffeinated carbonated drinks.**
- **Commonest flare self-management: drinking extra fluid** — whether as natural hydrodistension or dilution of irritants is unclear; also rest, dietary change, heat or cold, relaxation.

- **Urine pH hypothesis:** acid-sensing ion channels are expressed, and some upregulated, in IC/BPS bladders. But a **double-blind crossover instillation of urine at pH 5.0 vs 7.5 showed no difference in pain** → manipulating urine pH has little effect. Orange and grapefruit juices, rich in potassium and citrate, **raise urine pH**, yet patients avoid them — either triggers are misattributed or another process (e.g. **bladder-bowel cross-talk**) operates. Alkalinising is still worth trying — calcium glycerophosphate (Prelief, an over-the-counter food-acid reducer) or baking soda (sodium bicarbonate) — without controlled trials.
- **Elimination diet — sequence:** diary of food, voiding and pain → bland diet from compiled food lists (Box 28.4) → slowly reintroduce suspect items, recording the effect → avoid only confirmed triggers. Triggers are individual, some patients have none, and the method **prevents unnecessary restriction and nutritional deficit**. A short validated questionnaire captures baseline sensitivities.
- **Education and behavioural modification programme (EBMP)**, the control arm of the **landmark NIH amitriptyline RCT (Foster et al., 2010)**, in newly diagnosed patients, covered four areas: (1) symptom control, (2) fluid control, (3) diet, (4) **bladder training and urge suppression**, plus behavioural stress and pain management. **EBMP alone: 45% moderately or markedly improved at 12 weeks; adding amitriptyline (55%) was not significantly better.** Separately, hydrodistension followed by bladder training beat hydrodistension alone at 24 weeks.

BOX 28.4 FOODS/BEVERAGES IDENTIFIED AS MOST AND LEAST BOTHERSOME TO PATIENTS WITH IC/BPS

Category	Most bothersome	Least bothersome
Beverages	Coffee (caffeinated and decaffeinated), caffeinated tea; all carbonated drinks (cola, non-cola, diet, caffeine-free); beer, red and white wine, champagne; cranberry , grapefruit, orange, pineapple juices	Water ; milk (low-fat and whole)
Fruit	Grapefruit, lemon, orange, pineapple	Bananas, blueberries, honeydew melon, pears, raisins, watermelon
Vegetables	Tomato and tomato products	Broccoli, Brussels sprouts, cabbage, carrots, cauliflower, celery, cucumber, mushrooms, peas, radishes, squash, courgette, white and sweet potatoes/yams
Spice, condiments	Hot peppers, spicy foods, chilli, horseradish, vinegar, monosodium glutamate	—
Sweeteners	NutraSweet, Sweet'N Low, Equal, saccharin	—
Cuisines	Mexican, Thai, Indian	—
Protein	—	Eggs, turkey, beef, pork, lamb, shrimp, tuna, salmon, chicken
Grains, snacks	—	Oats, rice, pretzels, popcorn

Modified from Friedlander, Shorter and Moldwin (2012).

- **The pattern:** the offending list is **acid, caffeine, alcohol, carbonation, artificial sweeteners and spice**; the safe list is bland, non-acidic whole food — the starting point of the elimination diet.

The Role of Patient Support Groups

- **Interstitial Cystitis Association (ICA)**, not-for-profit since 1984: education for doctors and patients, symposia, charity walks, research promotion and its own young-investigator pilot research programme.

Also the Interstitial Cystitis Network (ICN) and the not-for-profit International Painful Bladder Foundation (IPBF). Virtual platforms widen access.

Oral Therapies

- **Few oral agents have unequivocal efficacy in large multicentre RCTs, and none is shown to alter natural history;** consider periodic “**drug holidays**” to check effect, and multimodal therapy to avoid high-dose single-drug toxicity.

COMPARISON: ORAL AGENTS

Agent	Dose	Evidence, bottom line	Key toxicity
Amitriptyline	10 mg nocte, +10 mg weekly → ~50 mg (range 25–75 mg)	RCT positive; NIH RCT negative overall but effective if ≥50 mg tolerated ; best in widespread pain; HL-independent	Sedation, anticholinergic effects; cardiac contraindications
Hydroxyzine	25 mg nocte → 50 mg nocte + 25 mg mane	NIH RCT negative; aids sleep; systemic-allergy subset	Sedation
Cimetidine (H2)	400 mg twice daily	One positive placebo RCT; mechanism unexplained	—
PPS	100 mg three times daily ; effect in 3–6 months	Conflicting; phase IV RCT negative ; less effective in HL	Pigmentary maculopathy
Ciclosporin A	1–3 mg/kg/day , divided	Best in refractory HL	Nephrotoxicity, hypertension
Sildenafil; tadalafil	25 mg daily; 5 mg daily	Small RCT; case series	—
Gabapentin; pregabalin	—	No randomised studies in IC/BPS	—
Quercetin	500 mg twice daily	Small uncontrolled	—
Antibiotics	—	No role without documented infection	Frequent adverse events

Amitriptyline

- **A TCA and a staple of oral treatment.** Properties: (1) central and peripheral anticholinergic action; (2) blockade of serotonin and noradrenaline reuptake; (3) sedation — central, but related also to (4) antihistamine action, amitriptyline being **one of the most potent H1-blocking TCAs**. It may also **stimulate β-adrenergic receptors in the bladder body**, reducing detrusor excitability and aiding storage.
- **Origin:** a “serendipitous” response in a patient treated for depression. The rationale — low-dose efficacy in chronic pain plus anticholinergic, β-adrenergic, sedative and H1 actions — led to trials at 25–75 mg with promising results but high dropout; **sedation the commonest complaint**. Uncontrolled studies (and desipramine) also reported success.
- **Double-blind intention-to-treat RCT (25–100 mg as tolerated):** good or excellent satisfaction far more often than placebo; **little tachyphylaxis at 19 months**, with response across the symptom spectrum.
- **Landmark NIH RCT (Foster et al., 2010):** EBMP with vs without amitriptyline 10–75 mg → **55% vs 45%, not significant**. Subgroup tolerating ≥50 mg: **77% vs 53%** — efficacy beyond EBMP.
- **Phenotype reanalysis:** benefit confined to **widespread pain**; no significant effect in localised pelvic pain.

- **Efficacy independent of HL status**; cystoscopy does not predict outcome. May also help accompanying pain disorders (e.g. vulvodynia).
- **Dosing sequence**: 10 mg at bedtime → increase by 10 mg weekly → maximum tolerated dose of 50 mg or more at bedtime. Median preferred dose 50 mg (range 25–75 mg); slow titration minimises side-effects, and **long-term benefit may occur below 50 mg**.
- **Side-effects**: fatigue, constipation, dry mouth, increased appetite and weight gain, loss of libido, nausea, dizziness → take at bedtime. **Delayed ejaculation, anorgasmia, reduced libido** are common to all TCAs.
- **Caution**: obstructive voiding or incomplete emptying and chronic constipation (both worsened); orthostatic hypotension.
- **Contraindications**: long QT syndrome; significant conduction disease (bifascicular or trifascicular block); myocardial infarction within 6 months; unstable angina; congestive heart failure; frequent premature ventricular contractions; history of sustained ventricular arrhythmia. **Doses ≥100 mg raise the relative risk of sudden cardiac death**.
- **Nortriptyline** (secondary amine) may cause fewer side-effects than amitriptyline (tertiary amine) and is used in other pain syndromes, but **no IC/BPS trials**.

Antihistamines

- **Both H1 and H2 receptors are overexpressed in IC/BPS bladders**; in animals, mast-cell-dependent pelvic pain is mediated by both.
- **Hydroxyzine**: rationale — blocks neuronal activation of mast cells. Uncontrolled, unblinded series: almost every symptom improved by about 30%; **better in patients with documented allergy or evidence of bladder mast-cell activation**. **NIH placebo RCT: no significant response**. Still useful for its safety and **sleep benefit (often with amitriptyline)**, especially in patients with **systemic allergies**.
- **Cimetidine**: two-thirds improved in uncontrolled series (600 mg daily, divided); effective in a double-blind placebo RCT at 400 mg twice daily. Mucosa histologically unchanged → **mechanism unexplained**; H1/H2 expression does not differ between responders and non-responders to cimetidine or amitriptyline. Widely used in the UK (over a third of patients).

Pentosan Polysulfate Sodium

- **Approved 1996 under the FDA orphan products programme — still the only FDA-approved oral drug for IC** (“relief of bladder pain or discomfort associated with interstitial cystitis”).
- Synthetic sulfated polysaccharide resembling bladder-surface GAGs. **Only 3–6% of an oral dose reaches the urine**. Proposed actions: **repair of GAG defects** in a “leaky” urothelium; sequestration of “toxic cations”. **May take 3–6 months to work**.

COMPARISON: PPS TRIALS

Trial design	Dose	Verdict
Initial series	—	Encouraging
Two parallel placebo RCTs	100 mg three times daily	PPS > placebo, low absolute response
Crossover placebo RCT	—	PPS > placebo
Two further placebo RCTs (including NIH)	200 mg twice daily; 100 mg three times daily	No difference

Trial design	Dose	Verdict
Phase IV placebo RCT (Nickel et al., 2015), intention-to-treat; endpoint $\geq 30\%$ fall in ICSI	100 mg once daily; 100 mg three times daily	No difference: 39.8% vs 40.7% placebo
Open-label RCT vs CyA	—	CyA far better

- **Phase IV caveats:** looser enrolment definition (NIDDK substratification performed); comorbid IBS, pelvic-floor dysfunction and depression not excluded; prior PPS users included; dropout for reasons other than non-response.
- **Overall response poor, about 30–40%, even where better than placebo;** conflicting evidence and heterogeneity plague the trials. Small studies suggest value in radiation and cyclophosphamide cystitis.
- **Pigmentary maculopathy:** unique retinal pigmentary change, first in case reports and then confirmed in retrospective studies. Difficulty reading, slow adjustment to low light, blurred vision, rarely blindness. **Risk rises with cumulative dose; changes may be irreversible and may progress after stopping.**
- **FDA label warnings (2020) — sequence:** (1) ophthalmological history in all before starting → (2) comprehensive baseline retinal examination before starting if pre-existing eye disease → (3) **baseline retinal examination for all within 6 months of starting, then periodically** → (4) if pigmentary change develops, re-weigh risk and benefit. Long-term users should be counselled before continuing.
- **Other adverse effects (<4%):** reversible alopecia, diarrhoea, nausea, rash; rare bleeding. Caution in high breast-cancer risk and premenopausal women (promotes MCF-7 breast-cancer cell proliferation in vitro). Tachyphylaxis uncommon.
- **Authors' view:** usefulness limited by conflicting efficacy, safety concern, inability to select responders (**may be less effective in HL disease**), high cost and slow onset; controversial. **Remains an AUA Option because it is the only FDA-approved oral drug.**

Immunoregulating agents

- **Chronic oral steroids are not recommended.** Studied and ineffective: suplatast tosilate, mycophenolate mofetil, adalimumab. CyA may help refractory HL disease (above).

Quercetin

- Over-the-counter bioflavonoid with possible anti-inflammatory action. **500 mg twice daily for 4 weeks:** almost all of a small uncontrolled group improved on O'Leary–Sant and global scores; placebo-controlled trials needed.

Antibiotics

- **Excluding UTI is integral to diagnosis.** No organism has been consistently identified by any technology; the urinary microbiome and metabolome may yet inform the question.
- **Double-blind placebo RCT:** 18 weeks of **rifampicin plus sequential doxycycline, erythromycin, metronidazole, clindamycin, amoxicillin and ciprofloxacin (3 weeks each)**. Improvement was more frequent with antibiotics but **not significant**, confounded by new IC/BPS therapy in both arms. **Adverse events were significant: 80% vs 40%**, mainly nausea/vomiting and diarrhoea. Most correctly guessed their arm, and correct guessers were more likely to report improvement; no durable benefit was reported.
- A large doxycycline series in women with urgency, frequency and urethral/pelvic pain reported high success, but in a **population broader than IC/BPS**; most IC/BPS patients have already received empirical antibiotics before diagnosis.

- **No evidence for antibiotics without culture-documented infection; AUA recommends against chronic antibiotics.** Incidental bacteriuria without a flare is unlikely to explain baseline symptoms.
- **One empirical course is reasonable if the patient has never had antibiotics for the urinary symptoms.** A flare thought infective → **urinalysis and culture mandatory.**
- **Recurrent uncomplicated bacterial cystitis** may trigger prolonged flares, with hypersensitivity persisting after bacteriological clearance (pain sensitisation) → consider **prophylaxis.**

Phosphodiesterase inhibitors

- **Phosphodiesterase type 5 inhibitors (PDE5Is)** may relax smooth muscle or afferent structures and suppress spontaneous smooth-muscle activity. A small double-blind placebo RCT of **sildenafil 25 mg daily** (women without HLs meeting NIDDK criteria) improved ICSI/ICPI, UDS parameters and VAS at 3 months — never followed up. A case series of **tadalafil 5 mg daily** improved frequency by 4 weeks and pain by 12 weeks.

Gabapentinoids

- **Gabapentin** (anticonvulsant) is effective in neuropathic pain — diabetic neuropathy, post-herpetic neuralgia — and synergistic with morphine; may benefit CPPS and IC/BPS. **Pregabalin** is effective in neuropathic pain and FMS. **No randomised studies of either in IC/BPS.**

TABLE 28.4 GRADE AND LEVEL OF EVIDENCE OF ORAL AND INTRAVESICAL THERAPIES FOR THE INTERNATIONAL CONSULTATION ON INCONTINENCE, THE EUROPEAN ASSOCIATION OF UROLOGY GUIDELINE, AND GIANNANTONI ET AL.

Treatment	ICI	EAU	Giannantoni	AUA 2022
Oral therapies				
Amitriptyline	B: 2	A: 1	A: 1	B, Option
PPS	D: 1	A: 1	C: 1	B, Option
Cimetidine	C: 3	—	—	B, Option
Ciclosporin A	C: 3	A: 1	A: 1	C, Option
Hydroxyzine	D: 1	A: 1	—	C, Option
Analgesics	C: 4	C: 2	—	—
L-arginine	—	A: 1	A: 1	—
Nalmefene; duloxetine	—	A: 1; C: 4	—	—
Antibiotic regimens	D: 4	—	—	B, should not be offered
Gabapentin	C: 4	—	—	—
Tanezumab	D: 1	—	—	—
Benzydamine; suplastat tosilate	D: 3	—	—	—
Azathioprine, chloroquine derivatives, doxycycline, methotrexate, misoprostol, montelukast, nifedipine, quercetin, vitamin E	D: 4	—	—	—
Intravesical therapies				
Lidocaine	C: 2	—	—	B, Option
DMSO	B: 2	A: 1	—	C, Option

Treatment	ICI	EAU	Giannantoni	AUA 2022
Heparin	C: 3	—	—	C, Option
BTX (intramural)	A: 1	—	A: 1	C, Option
Hyaluronic acid	D: 1	B: 2	—	—
Chondroitin sulfate	D: 4	B: 2	A: 1	—
PPS (intravesical)	D: 4	A: 1	—	—
Oxybutynin	D: 4	—	—	—

International Consultation on Incontinence (ICI; Hanno et al., 2013); European Association of Urology (EAU) guideline on chronic pelvic pain (Fall et al., 2010); Giannantoni et al. systematic review (2012); AUA (Clemens et al., 2022). Modified from the Committee on Bladder Pain Syndrome, Fifth International Consultation on Incontinence (2012). Oxford grades. AUA evidence strength: **A** well-conducted RCTs or exceptionally strong observational studies; **B** RCTs with some weaknesses or generally strong observational studies; **C** inconsistent, small or confounded observational studies. **Option** = non-directive statement leaving the decision to clinician and patient, because benefit and burden appear balanced or unclear.

- **What the table tests:** the bodies disagree — PPS is graded D by ICI but A by EAU — mirroring the conflicting trials. **Evidence grade is not recommendation strength:** the AUA makes almost every drug an Option.
- **Trap:** antibiotics are the one agent the AUA says should not be offered.

Concept — No oral drug works in all-comers, so each is matched to a phenotype: amitriptyline to centralised pain, antihistamines to the allergic and localised subset, CyA to the inflammatory HL lesion — while PPS's GAG-repair rationale has not translated into consistent benefit.

Intravesical Therapies

- **Appeal:** treats the end-organ directly without systemic side-effects; suits patients intolerant of oral agents, or whose response to oral and conservative measures has plateaued; **best suited to bladder-centric pain** — e.g. relief with a trial of intravesical lidocaine.
- **Downside:** pain generated centrally or by other pelvic generators (e.g. HTPFD) is left untreated; response is limited by poor adherence and poor absorption across the urothelium.
- **Caustic agents** (silver nitrate, oxychlorosene sodium [Clorpactin]) are rarely used now. Many current agents can be **self-instilled**, empowering patients and reducing visits.
- **AUA:** intravesical instillation, alone or combined, is an Option. Intravesical lidocaine is a good option for flares because its effect is immediate.

COMPARISON: INTRAVESICAL AGENTS

Agent	Typical regimen	Evidence, bottom line
DMSO 50%	50 mL, 15–20-min dwell, every 1–2 weeks for 4–6	Two randomised crossover trials positive; recent placebo RCT positive in bladder-centric (mostly HL) patients; not in non-HL
Heparin	10,000–40,000 units (e.g. 40,000 U daily, self-instilled, held 30–60 min)	Uncontrolled positive; effect may take 6 months–2 years
PPS (intravesical)	300 mg twice weekly	No benefit over oral PPS (systematic review)
Hyaluronic acid	40 mg weekly for 4–10 weeks, then monthly	Uncontrolled positive; two placebo RCTs negative (unpublished) ; not US-approved
Chondroitin sulfate	2% weekly ×6, then monthly ×4	Open-label positive; larger study negative

Agent	Typical regimen	Evidence, bottom line
Hyaluronic acid + chondroitin	—	Pain reduction similar to DMSO, fewer adverse events (open-label RCT)
Alkalinised lidocaine	200 mg + 8.4% bicarbonate (10 mL) daily ×5	RCT: significant but modest benefit at day 3
Heparin + alkalinised lidocaine	40,000 U + 8 mL 1–2% lidocaine + 3 mL 8.4% bicarbonate	High success in open-label; better than lidocaine alone (pilot)
Lidocaine depot (LIRIS)	400 mg intravesical device	Two RCTs negative, HL and non-HL
Triamcinolone (added)	40 mg in cocktail	No added benefit (RCT; HL status unreported)

Dimethyl Sulfoxide

- **50% DMSO is FDA-approved only as a stand-alone treatment**, though often given in a “cocktail” with sodium bicarbonate, heparin, steroids and/or lidocaine.
- **By-product of wood pulp, a lignin derivative**; exceptional solvent, miscible with water, lipids and organic agents. Properties: membrane penetration, enhanced drug absorption, anti-inflammatory and analgesic action, collagen dissolution, muscle relaxation, mast-cell histamine release. In-vitro effects belie in-vivo benefit, and **no histamine release is seen after treatment**; it may **desensitise lower-urinary-tract nociceptive pathways**. Human use began in the 1960s (musculoskeletal inflammation, scleroderma skin).
- **Two randomised crossover trials**: DMSO vs saline (4 instillations of 50 mL at 2-week intervals, 15-min dwell) and DMSO vs bacille Calmette–Guérin (BCG, ineffective in IC/BPS; 6 weekly instillations) both favoured DMSO.
- **Patients with DO do not respond; HL patients may respond better** — pain improves with or without HLs, frequency only with HLs. A multicentre double-blind placebo RCT in Japanese patients with bladder-centric features (mostly HL, lidocaine responders; 50 mL every 2 weeks for 12 weeks, 15-min dwell) improved ICSI; **post hoc, no efficacy in non-HL** → anti-inflammatory action suits inflammatory lesions.
- **Practical**: easy, low morbidity. Rat bladder strips: contraction abolished at 40%, compliance reduced at 30%, negligible effect at ≤25% (human relevance unknown). Rare eosinophilic cystitis after DMSO. **Warn of severe urgency and frequency with the first instillations. Dwell 15–20 minutes** — rapidly absorbed; longer holding is unnecessary and may worsen pain. **Garlic-like breath** (partial pulmonary excretion) makes blinded trials impossible.
- **Cocktail**: 50 mL Rimso-50 + 80 mg triamcinolone + 44 mEq sodium bicarbonate + 20,000–40,000 units heparin ± lidocaine, **weekly for 6 weeks**; responders may have **monthly maintenance for 6 months**. **DMSO enhances absorption of co-administered drugs → risk of systemic lidocaine toxicity**. Cohorts report about 60% success and propose first-line use; no combination has proved superior to DMSO alone (systematic review).

Glycosaminoglycans (Heparin, Pentosan Polysulfate, Hyaluronic Acid, Chondroitin Sulfate)

- **Exogenous GAGs restore the epithelial permeability barrier** after protamine injury.
- **Heparin** mimics the bladder’s mucopolysaccharide lining; anti-inflammatory; inhibits fibroblast proliferation, angiogenesis and smooth-muscle proliferation; **virtually no systemic absorption, even from an inflamed bladder**. Regimens: 10,000 U in sterile water alone or with DMSO; 25,000 U twice weekly for 3 months (≥50% improvement in International Prostate Symptom Score in most women); 40,000 U in 20 mL daily, self-administered and held 30–60 min, with **“reasonable improvement” expected between 6 months and 2 years**.

- **Adding alkalinised lidocaine improves pain relief** (e.g. 8 mL 2% lidocaine + 4 mL 8.4% bicarbonate). Japanese regimen: 20,000 U heparin + 5 mL 4% lidocaine + 25 mL 7% bicarbonate weekly for 12 weeks — high success, most still improved at 1 month after the last instillation, about half at 2 months. **Alkalinised lidocaine–heparin is proposed for flares.**
- **Intravesical PPS:** modest benefit in a small trial; oral + intravesical no better than oral alone; **systematic review: no additional benefit.**
- **Hyaluronic acid** (non-sulfated GAG): 40 mg in 46 mL saline weekly for 4–6 weeks, then monthly; uncontrolled response rates vary widely. **Two industry multicentre double-blind placebo RCTs (2003, 2004) were negative and never published; not approved in the US.** Best results in KCl-positive patients able to hold the solution 2 hours, weekly for ≥10 weeks; refractory cases treated with hydrodistension plus instillation every 1–3 months. **Efficacy unproven in controlled, blinded trials;** frequency and duration of treatment vary widely.
- **Chondroitin sulfate:** important for barrier function; **luminal deficit shown in IC by immunohistochemistry;** inhibited inflammatory-cell recruitment in a “leaky bladder” model. Small uncontrolled studies show variable success; a multicentre open-label study (2% weekly for 6 weeks, then monthly for 4 months) found most responded without safety issues, but **a larger follow-up study failed to show significant efficacy.** Open-label use in “chronic cystitis” (up to 8 weekly instillations) improved urgency, voided volume and nocturia.
- **Combined GAGs:** favourable in uncontrolled series; an open-label RCT found hyaluronic acid + chondroitin sulfate matched 50% DMSO for pain with fewer adverse events.
- **GAG replenishment overall: two decades of use, mostly small, poor, uncontrolled studies** → large RCTs in well-phenotyped groups are urgently needed.

Sodium channel blocking anaesthetics (lidocaine)

- **Diagnostic and therapeutic; almost immediate effect** → useful for flares or maintenance. A positive response to alkalinised lidocaine is used by many centres as **evidence of a bladder-centric phenotype.**
- Lidocaine: sodium-channel blocker, also anti-inflammatory and antimicrobial; short serum half-life; **not absorbed through urothelium to a significant degree unless alkaline** → **alkalinise with sodium bicarbonate.**
- **RCT of 5 consecutive daily alkalinised lidocaine instillations** (10 mL containing 200 mg lidocaine and 8.4% bicarbonate) vs placebo: moderate or marked improvement by day 3 in 30% vs about 10%. **The modest 30% response underlines how often pain is extravescical.** In the open-label phase about half improved at 3 and 10 days. Alkalinised and non-alkalinised lidocaine have never been compared.
- **Heparin + alkalinised lidocaine:** 40,000 U heparin + 8 mL 1% lidocaine + 3 mL 8.4% bicarbonate gave high open-label success, higher still with 2% lidocaine. A single-blind pilot found heparin–lidocaine better than alkalinised lidocaine alone for bladder pain, urgency and GRA at 1 and 24 hours; serum absorption exceeded that of non-alkalinised lidocaine.
- **Extended-release lidocaine-releasing intravesical system (LiRIS), 400 mg vs placebo device: two multicentre RCTs negative in both HL and non-HL patients.**
- **Bupivacaine** suggested, but no controlled trials. **Lidocaine may cause tachyphylaxis** — **monitor for waning efficacy.**

Intravesical triamcinolone

- RCT adding **40 mg triamcinolone** to a heparin–lidocaine–bicarbonate–bupivacaine cocktail: after six instillations, O’Leary–Sant improvement did not differ → **no added benefit;** HL status was not reported, so applicability to HL disease is unknown.

BOX 28.5 CLINICAL PEARLS FOR INTRAVESICAL INSTILLATION

- **Authors' office cocktail:** 20 mL 2% lidocaine jelly (400 mg) + 15 mL 0.5% bupivacaine (75 mg) + 20,000 units heparin + gentamicin 80 mg/2 mL; **optional triamcinolone 80 mg/2 mL for HL patients.** The anaesthetic component may be replaced by 20 mL 2% lidocaine (400 mg) + 10 mL 8.4% sodium bicarbonate.
- **Sequence:** ask pain intensity before → instil through a small catheter → observe for adverse reactions (e.g. perioral numbness) in the office for the first few instillations → **if a reaction occurs, drain immediately and wash out with sterile water or saline** → ask pain after instillation and record whether it improved.
- **The optimal agent, combination, dose, schedule and value of maintenance are not established;** cocktails have never been compared directly.

Concept — *Intravesical therapy can only treat pain that arises in the bladder, so the anaesthetic challenge is both the diagnostic test and the selection criterion: a poor lidocaine response predicts poor results from every instillation.*

Intradetrusor Therapy**OnabotulinumtoxinA**

- **FDA-approved for refractory neurogenic DO and OAB;** long used for muscular hypercontraction and described for HTPFD and vulvodynia. **Not FDA-approved for IC/BPS,** but observational studies, a few RCTs and several meta-analyses support efficacy and low risk in selected cases. **AUA Option.** Repeated injections, at intervals or on recurrence, remain effective; earlier use may follow shared decision-making. **Best suited to bladder-centric patients.**
- **Mechanisms beyond inhibiting presynaptic acetylcholine release:** analgesic (not merely relief of spasm); **reduces peripheral sensitisation** by inhibiting glutamate and substance P release and c-Fos expression; reduces sensory feedback from muscle (acetylcholine release from gamma motor neurons to spindle fibres); inhibits sensory neurotransmitter release in rat models of acute injury and chronic inflammation; **reduces chronic inflammation and apoptosis** after repeated injection in IC/BPS; blocks calcitonin gene-related peptide (CGRP) release in rats; restores afferent responses and minimises DO after barrier disruption in animals.

COMPARISON: BTX-A TRIALS

Design	Template	Finding
200 U vs 100 U vs hydrodistension alone (repeat hydrodistension at 2 weeks in all)	Trigone-sparing; posterior and lateral walls, 40 sites	Both doses beat hydrodistension at 3 months; 200 U unacceptable (dysuria, raised PVR/retention) → 100 U standard; a few at 100 U had raised PVR or retention
100 U vs saline, both followed by hydrodistension	Trigone-sparing, 20 sites	Greater pain reduction with BTX-A at 2 months
100 U trigonal (uncontrolled)	10 trigonal sites	Efficacy in over half, lasting ~9 months; no voiding dysfunction
100 U trigonal vs saline, no hydrodistension	10 sites, medial to ureteric orifices	Greater pain reduction; most achieved ≥50% pain reduction; no retention
Immediate vs delayed (1 month) 100 U trigonal	30 trigonal sites	Better GRA at 1 month; response lasts about 5 months

Design	Template	Finding
Trigone-including vs trigone-sparing	—	No difference in efficacy or complications (UTI, retention)

- **Why the trigone:** sensory nerves are densest there; limiting injection to the trigone maximises sensory effect while sparing the detrusor body, lowering retention risk.
- **HL vs non-HL response: underpowered and conflicting.** Two small trigonal studies found no difference; a large Taiwanese multivariate analysis found **HL status not predictive, but maximum capacity ≥ 760 mL at hydrodistension predicted response** — contradicting an earlier study from the same group with poor HL response.
- **BTX-A cannot cross the urothelium → must be injected; liposome-encapsulated intravesical BTX-A was no better than placebo.**

BOX 28.6 CLINICAL PEARLS OF BTX-A INJECTION

- **Indication:** IC/BPS refractory to conservative, oral and intravesical treatment; AUA Option; shared decision-making, including that BTX-A is **not FDA-approved** for IC/BPS; experienced clinicians only.
- **Technique:** 100 U diluted in 5 mL preservative-free saline (20 U/mL), injected into the trigone in 1-mL aliquots, limiting volume load and minimising impaired emptying.
- **Prerequisite:** willing and able to perform clean intermittent catheterisation (CIC). Retention risk is low but not zero, and CIC is hard for IC/BPS patients — urethral pain from catheterisation, and frequent catheterisation because of low functional capacity and hypersensitivity. Patients already self-catheterising for instillations are good candidates. **Use with caution when symptoms or signs of incomplete emptying are present.**

Neuromodulation

- **International Neuromodulation Society definition:** “the alteration of nerve activity through targeted delivery of a stimulus, such as electrical stimulation or chemical agents, to specific neurologic sites in the body.” Appeal: modulates nociceptive transmission and pain perception; mechanism poorly understood (**gate-control theory**); may reduce pelvic-floor hypertonicity and HTPFD symptoms.

COMPARISON: NEUROMODULATION MODALITIES

	Transcutaneous electrical nerve stimulation (TENS)	Posterior tibial nerve stimulation (PTNS)	Sacral neuromodulation (SNM)
Mechanism	Activates large-diameter non-noxious A β afferents (pain gate); endorphin release into cerebrospinal fluid	Peripheral afferent stimulation	Pain-gate modulation at spinal and higher levels; endorphins; pelvic-floor modulation
FDA-approved for	—	OAB	Non-obstructive retention; OAB (UUI, urgency-frequency); faecal incontinence — not IC/BPS
IC/BPS evidence	Open-label; better with HLs than without ; no sham control	Placebo-controlled negative; uncontrolled weak	Uncontrolled; long-term pain reduction; works with and without HLs
Status	Adjunct; RCTs needed	Role questionable	AUA Option (with BTX-A)

	Transcutaneous electrical nerve stimulation (TENS)	Posterior tibial nerve stimulation (PTNS)	Sacral neuromodulation (SNM)
Caveat	Electrode site and settings undefined	Strong placebo effect	Higher adverse events, recurrence and explantation in IC/BPS

- **TENS:** the simplest neuromodulation; may also reduce frequency. In the most complete review, suprapubic TENS produced favourable results or remission in about a quarter of non-HL and — surprisingly — about half of HL patients. Caveats: open-label, small numbers, no sham group, and a **significant placebo effect of peripheral stimulation**. **Advantages: easily combined with other conservative strategies; non-invasive and reversible.**
- **PTNS:** little evidence for IC/BPS pain. A double-blind placebo RCT of transdermal laser stimulation showed no benefit but a **remarkably strong placebo effect** (showing why placebo arms matter); twice-weekly and weekly courses showed no clinically relevant improvement; an uncontrolled weekly course improved pain, frequency and urgency but not nocturia; another uncontrolled study showed poor GRA response.
- **SNM (usually S3): Patients with identifiable pain and HTPFD do best;** pelvic pain without demonstrable pelvic-floor dysfunction or levator tenderness does poorly. Response associated with falling urinary chemokines (notably monocyte chemoattractant protein-1 [MCP-1]); successful test stimulation lowered antiproliferative activity and normalised heparin-binding epidermal growth factor-like growth factor (HB-EGF).
- **SNM evidence:** from the late 1990s; improvement in most patients with intractable pelvic pain and in about two-thirds of IC/BPS; a single-blind randomised crossover trial suggested **pubendal stimulation may outperform sacral**. Largely uncontrolled series show **long-term pain VAS reductions of 41–63%**, with efficacy **with and without HLs** → AUA Option alongside BTX-A.
- **Technique: staged permanent quadripolar lead (1–4 weeks) gives higher implantation rates than a 3–4-day percutaneous nerve evaluation (PNE)**, with similar long-term efficacy once implanted; newer PNE leads displace less. **Unilateral = bilateral stimulation** in the only randomised crossover trial.
- **Cautions: pain predicts adverse events;** SNM helps over half with urgency–frequency, but caution when pain dominates. **IC/BPS predicts early generator removal (odds ratio [OR] 10.5);** among SNM indications, IC/BPS has the most long-term adverse effects (about a quarter) and more recurrence → explain likely problems before implantation. **FMS patients may not tolerate the generator;** whether smaller rechargeable generators are better tolerated is unstudied.
- **Other observations:** opioid requirement falls, but most chronic opioid users continue; results age-independent; female sexual function may improve; **urgency may predict long-term success;** frequent reprogramming needed. Dorsal root ganglion, spinal cord and deep brain stimulation may play a future role (case-level evidence only, mostly non-IC/BPS).

Choosing between	BTX-A	SNM
Favoured when	Bladder-centric pain; able and willing to perform CIC	Obstructive voiding, incomplete emptying, HTPFD-related pain, or CIC impossible or refused
Main risk	Retention requiring CIC	Adverse events, explantation

Concept — Both BTX-A and SNM are borrowed from OAB and work in IC/BPS mainly through sensory rather than motor effects; the choice between them is set by bladder emptying — BTX-A risks retention and demands CIC, whereas SNM suits obstructive voiding and pelvic-floor pain.

Surgical Therapy

Hydrodistention

- **AUA Option: short-duration (1–3 minutes), low-pressure (60–80 cm H₂O) hydrodistension under anaesthesia.**

BOX 28.7 CLINICAL PEARLS OF HYDRODISTENTION

- **Sequence:** initial cystoscopy (usually unremarkable) → if no HLs, distend **1–3 minutes at 80 cm H₂O by gravity**, inspecting throughout (mucosal or deeper tears occur rapidly); compress the urethra digitally against the scope if leakage occurs → empty and refill → observe for glomerulations → further therapeutic distension for several minutes → biopsy rarely needed, and **only after distension** (perforation) → instil anaesthetic at the end (20–30 mL 2% lidocaine, or 1:1 20 mL 1% lidocaine + 20 mL 0.25% bupivacaine).
- **Counsel that a post-procedure flare is common**, often needing a short course of urinary anaesthetics or oral analgesics.
- **Outcomes:** in a retrospective series, adding a transvaginal trigonal nerve block gave no extra benefit, but **pain scores fell by about half at 1 month**; 2-minute and ≥5-minute distensions were equivalent.
- About half to 60% respond, but **most responses last under 2 months; efficacy declines steeply — about half at 1 month, a quarter at 3 months, almost none (0–7%) at 6 months.**
- **Repeated hydrodistension does not reduce capacity**, with or without HLs.
- **Anaesthetic capacity <600 mL → better response** than larger capacities.
- Favourable results reported combining hydrodistension with BTX-A, with GAG therapy, and in ketamine cystitis. Capacity <400 mL marks a distinct bladder-centric phenotype (see Mapping Phenotypes).

Major Surgery

- **Primum non nocere applies here above all.** Cystoscopic triamcinolone/fulguration for HLs earns a strong AUA recommendation because of its high success and low complication rates; **major reconstruction only when all conservative treatment has failed.** Worst case: significant complications and continuing pain. Conversely, suicides have been attributed to unremitting pain when aggressive intervention was withheld.

Patient selection for major surgery

- **Major surgery (urinary diversion ± cystectomy, augmentation cystoplasty) is the last resort for rare, extremely carefully selected patients with bladder-centric symptoms who have failed all other therapy.**
- Risks: operative morbidity, mortality, devastating complications and irreversible lifestyle change (urostomy). **Pain relief is not guaranteed and may persist even after cystectomy**; the joy of symptom resolution is real, but evidence is small, heterogeneous retrospective series. Deliberate informed consent; if expectations are unrealistic, counsel further.
- Reserve for the motivated, well-informed patient with **extremely severe unresponsive disease — fewer than 10% of IC/BPS**; only experienced surgeons committed to IC/BPS care.
- **Sequence before considering surgery:** careful history → body map → pelvic-floor examination for levator tenderness/HTPFD → **trial of intravesical local anaesthetic** (profound relief localises pain to the bladder; an equivocal response suggests an extravesical source) → identify and treat other pelvic pain generators.

Predicts success (bladder-centric)	Predicts failure
HLs, especially refractory	No HLs

Predicts success (bladder-centric)	Predicts failure
End-stage fibrotic bladder; small capacity under anaesthesia	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal or no anaesthetic response
Painful filling, painful urgency, pain reproduced by touching mucosa with the scope	Urethral pain as the primary complaint; neuropathic pain
—	Other major pain sources: pelvic floor, widespread pain, polysymptomatic polysyndromic (PSPS) presentation with widespread somatic symptoms

- **HTPFD is the commonest extravescical finding, but may itself be a consequence of bladder pain.**
- **Differential spinal block:** if bladder pain persists after spinal anaesthesia to T10, the pain signal is generated higher in the cord and **bladder surgery will not relieve it.**
- **End-stage fibrotic bladder with intractable frequency and urgency but no pain → simple conduit diversion** may improve QoL regardless of pain.

Cystectomy or not?

- **Diversion without cystectomy, or trigone-retaining augmentation, is controversial:** bladder preservation lowers operative morbidity, but persistent bladder or trigonal pain may force later cystectomy or revision.
- **Diversion ± cystourethrectomy is the ultimate surgical answer — cutting the “Gordian knot”:** it eliminates urine–mucosa contact and bladder distension, and with them the pain, frequency and urgency. Unlike augmentation, **no incomplete emptying or CIC** (painful in IC/BPS) and lower metabolic risk.
- **Bladder left in situ:** lower morbidity, but future **persistent pain, pyocystis, haemorrhage, unremitting sense of incomplete emptying and spasm.** Bladder carcinoma has been reported but is not specifically associated with IC/BPS.

COMPARISON: MAJOR SURGICAL OPTIONS

Procedure	Outcome, bottom line	Selection point
Ileal conduit, bladder left in situ	About a fifth later need cystectomy , mostly for persistent pain, some for pyocystis; some remain in pain after cystectomy — all without HLs, one suicide	Capacity <200 mL under anaesthesia → better pain resolution
Primary cystectomy + ileal conduit	About three-quarters pain-free; all failures were non-HL ; most HL patients satisfied and would choose it again	HL
Cystectomy + continent diversion (e.g. Florida pouch, Kock pouch)	Variable: good results in some series, high failure with persisting pelvic and pouch pain in others; pouch pain in 40–50% within 6–36 months	Capacity <400 mL and no neuropathic pain predicted success; painful pouch can safely be retubularised into a conduit
Supratrigonal cystectomy + augmentation (ileum and/or caecum to a 1-cm trigonal cuff)	Mixed; overall failure about a quarter; about a sixth need completion cystectomy and diversion (some still in pain); most void spontaneously, but up to about 30% need self-catheterisation ; mucus irrigation often needed	Predictors: HLs; small anaesthetic capacity (<200–350 mL); avoid with urethral pain

- **Augmentation or continent reconstruction attracts the young, healthy patient** (no external appliance, no anastomotic stricture), but brings metabolic risk, incomplete emptying, painful self-catheterisation,

and **persistent pain or recurrent HL at the retained trigonal edge** → prefer the simplest diversion possible.

- **Recurring theme:** extremely careful selection and realistic expectations; surgery only by experienced surgeons committed to aftercare. Nothing is more tragic than irreversible surgery followed by continued pain, complications and a surgeon with nothing more to offer.

Concept — *Surgery removes the bladder, so it can only cure **bladder-generated pain**: every predictor of success (HL, small fibrotic capacity, anaesthetic response) marks bladder-centric disease, and every predictor of failure (urethral, neuropathic or widespread pain, large capacity) marks pain that will survive cystectomy.*

FINAL THOUGHTS AND FUTURE DIRECTIONS

- The chapter incorporates the **2022 AUA Guideline** and, beyond it, a **phenotypic approach**. Analogy: cardiologists do not treat all chest pain alike — they distinguish coronary disease, valvular disease, aortic aneurysm, cardiomyopathy, arrhythmia and pleuritic pain. **Urologists should not treat all bladder/pelvic pain alike.**
- **Bladder-centric, pelvic-floor-centric and widespread-pain subgroups** each likely benefit from tailored treatment — a step towards precision medicine; trials by phenotype are needed.
- **HL status has such bearing on care that it justifies formally separating the nomenclature: IC (with HLs) vs BPS (without).**
- Success also depends on the learned art of the field: medical detective work across many pain generators, and a plan built on the whole clinical picture.